Enhancing Long-Horizon Robot Manipulation with Hierarchical Task Decomposition and Subtask Finetuning

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***Abstract*—In long-horizon manipulation tasks, end-to-end Visual-Language-Action (VLA) policies often fail not due to perception errors, but because early execution mistakes propagate and destabilize later stages. Motivated by recent hierarchical VLA approaches, we study the effect of explicit task decomposition and subtask-specific finetuning on execution stability. Specifically, we adopt a two-level hierarchical formulation that conditions a low-level VLA policy on oracle subtask prompts, and empirically analyze its behavior in the RoboTwin simulation environment. Across 500 independent evaluation episodes, hierarchical conditioning improves the task success rate from 45.4% (227/500) to 53.8% (269/500) on a representative long-horizon task. Ablation results suggest that both explicit task decomposition and subtask-level finetuning contribute to improved execution consistency.**

***Index Terms*—Visual-language-action, Robotic Control Systems, Multi-step Decision Making, Subtask Finetuning**

I. Introduction

In recent years, visual-language-action (VLA) models have demonstrated strong potential for tackling complex robotic manipulation tasks. Despite these successes, existing end-to-end VLA models remain fundamentally limited. In particular, the absence of explicit intermediate reasoning structures forces such models to rely primarily on pattern fitting from training data. Consequently, execution-time errors are difficult to detect and recover from, leading to cascading failures that propagate across time steps and significantly degrade overall task performance.

To better understand these limitations, we study a two-level hierarchical VLA formulation that decomposes long-horizon manipulation tasks into a sequence of smaller subtasks. Rather than introducing a new architectural design, we adopt a commonly used hierarchical prompting strategy and empirically analyze its impact on execution stability. In this formulation, a high-level planner provides structured subtask prompts that condition a low-level action policy to execute the corresponding atomic actions. This separation allows us to examine how explicit subtask representations influence error propagation and execution robustness under a fixed VLA backbone.

By introducing oracle-defined subtask representations, this study enables a more fine-grained analysis of individual task phases and their contribution to overall performance. Experimental results show that hierarchical conditioning can lead to consistent improvements in task success rate and execution stability, particularly for tasks with clear temporal structure.

II. Related Work

A. Visual-Language-Action Models for Robot Manipulation

Visual-Language-Action (VLA) models aim to unify visual perception, language understanding, and action generation within a single policy framework, enabling robots to perform manipulation tasks conditioned on natural language instructions and visual observations. Typically, a VLA model takes the current visual observation together with a task-level language prompt as input, and directly outputs low-level control actions or discrete action tokens in an end-to-end manner.

Recent works have demonstrated the effectiveness of this paradigm across a variety of robotic manipulation benchmarks. For instance, the π_0 model adopts an end-to-end VLA architecture that directly maps visual observations and language instructions to robot actions, achieving competitive performance on diverse manipulation tasks [3]. Such models highlight the potential of large-scale multimodal pretraining for general-purpose robotic control.

Despite these successes, most existing VLA models rely on flat, single-policy architectures. In long-horizon or multi-stage manipulation tasks, this design often leads to unstable execution, as errors introduced at early stages cannot be explicitly corrected or reasoned about at later steps. This limitation has motivated growing interest in moving beyond purely end-to-end VLA formulations toward more structured decision-making mechanisms.

B. Long-Horizon Robot Manipulation and Planning

Long-horizon robot manipulation tasks require an agent to reason over extended temporal horizons, often involving multiple interdependent subtasks, delayed rewards, and strong state dependencies. In such settings, even small execution errors may accumulate over time and propagate to later stages, ultimately leading to task failure. These challenges have been extensively studied in both classical robotics and learning-based control literature [7].

To address these difficulties, prior work has explored a range of planning-based and hybrid approaches, including task-and-motion planning, receding-horizon control, and hierarchical reinforcement learning [8], [9]. More recently, learning-based methods have incorporated memory mechanisms, world models, or explicit planning modules to improve long-horizon reasoning and credit assignment [10], [11]. While these approaches enhance robustness and temporal abstraction, they often rely on specialized architectures or task-specific engineering assumptions.

Within the VLA paradigm, the long-horizon challenge becomes particularly pronounced. Most existing VLA policies directly map observations to actions without explicitly representing intermediate task structure. Consequently, their performance tends to degrade significantly on multi-step tasks that require consistent execution across stages, underscoring the need for introducing structured decision-making mechanisms into VLA models.

C. Hierarchical Visual-Language-Action Models

To overcome the limitations of flat VLA models in multi-step and state-dependent tasks, recent research has begun exploring hierarchical decision-making mechanisms inspired by structured reasoning in large language models. In particular, the success of Chain-of-Thought (CoT) prompting in improving multi-step reasoning in language models has motivated analogous explorations in embodied intelligence [12].

Hierarchical VLA models introduce intermediate representations between task-level language instructions and low-level actions, explicitly constructing a multi-level decision hierarchy. This hierarchy typically consists of high-level task decomposition, intermediate subgoal planning, and low-level action execution. By exposing intermediate reasoning processes, such models aim to improve execution stability, interpretability, and generalization in long-horizon manipulation tasks.

Existing hierarchical VLA approaches can be broadly categorized into two classes.

a) Hierarchical Prompting.: This class of methods introduces an additional high-level planner, typically implemented as a vision-language model (VLM), which generates a sequence of subtask instructions or subgoals. A lower-level VLA policy then executes these instructions. Representative works such as HiRobot adopt this design, where the high-level VLM performs task decomposition and the low-level controller focuses on subtask execution [6]. Hierarchical prompting offers strong modularity and powerful high-level reasoning capabilities. However, it typically relies on multiple interacting models and assumes that each generated subtask can be executed reliably, making it sensitive to execution errors at the subtask level.

b) Internal Hierarchical Modeling.: An alternative line of work embeds hierarchical reasoning directly within a single VLA model by designing multi-stage output structures. In this setting, the model explicitly generates intermediate plans, subtask representations, or structured action sequences before producing final control commands. Examples include Wall-OSS [5] and $\pi_{0.5}$ [4], which introduce explicit intermediate representations to guide long-horizon execution. These approaches preserve the benefits of end-to-end training while incorporating structured reasoning. However, existing methods often emphasize planning or representation learning, with relatively limited focus on optimizing subtask-level execution policies.

In contrast to these works, our study does not propose a new hierarchical architecture or planning mechanism. Instead, we adopt an existing hierarchical prompting formulation and focus on analyzing its empirical impact on execution robustness under controlled settings.

III. Method

In this section, we describe the hierarchical VLA formulation adopted in our experiments. The goal is not to introduce a new model architecture, but to provide a clear description of the hierarchical setup used to analyze execution behavior in long-horizon manipulation tasks.

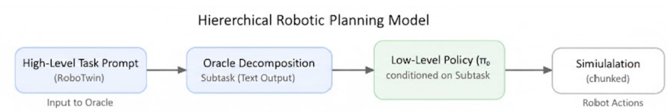


Fig. 1: Overview of the hierarchical VLA setup used in our experiments.

A. Overview of the Two-Level Hierarchical VLA Framework

We adopt a two-level hierarchy consisting of a high-level subtask planning layer and a low-level action execution layer. This formulation follows common practice in prior hierarchical VLA work and is intentionally kept minimal to isolate the effect of subtask conditioning.

The high-level layer determines what subtask should be executed next, while the low-level layer focuses on how to execute the corresponding atomic actions. Communication between the two layers is mediated through structured, subtask-specific prompts, which serve as a lightweight interface without modifying the underlying VLA architecture.

B. High-Level Subtask Planning

In our experiments, subtask decomposition is provided by an oracle using manually specified rules derived from task semantics. This design choice allows us to focus on analyzing the effects of explicit task structure, without introducing additional uncertainty from learned planners.

The high-level planner outputs a subtask identifier along with a corresponding language prompt specifying the current objective. Importantly, the planner does not directly generate actions or trajectories. Instead, it provides a structured context that conditions the low-level policy, ensuring that action selection is guided by the current stage of the task rather than by a fixed global trajectory.

Subtasks are executed sequentially, and the planner advances only after the current subtask terminates. While the planner does not incorporate explicit recovery mechanisms, this sequential structure introduces a weak form of temporal abstraction that helps limit the propagation of early-stage errors.

C. Low-Level Action Execution

The low-level action execution layer is implemented using an existing VLA policy, which takes as input the current visual observation together with the subtask-specific prompt from the high-level layer, and outputs atomic actions.

Crucially, the low-level policy architecture is not modified in our study. By varying only the subtask prompt while keeping the policy fixed, the model is encouraged to learn reusable, subtask-aligned action primitives.

For example, when executing a grasp-related subtask, the policy focuses on perception-action mappings relevant to grasping, whereas navigation or placement behaviors are implicitly de-emphasized. This selective activation emerges from prompt conditioning rather than explicit action masking or rule-based control.

D. Subtask-Specific Finetuning

To evaluate the effect of subtask-specific finetuning, we further finetune the same low-level VLA policy using data annotated with subtask prompts. This setting allows us to study how subtask-aligned supervision influences execution consistency, without introducing additional model capacity.

During finetuning, the input prompt includes both the overall task description and the current subtask instruction. We intentionally exclude explicit success criteria or environment-specific heuristics, focusing solely on strengthening the association between subtask semantics and appropriate action patterns. This choice helps prevent overfitting to narrow task definitions and encourages more generalizable subtask execution.

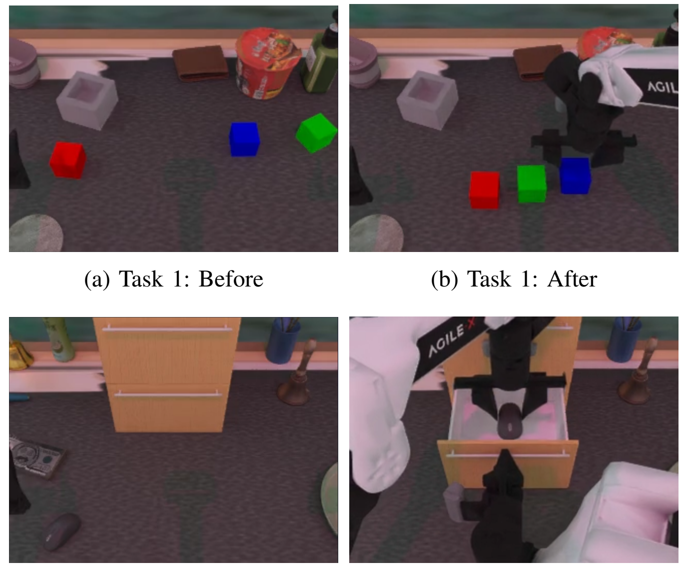
As a result, the finetuned policy exhibits improved consistency within individual subtasks and reduced sensitivity to execution deviations, which collectively provides evidence that structured subtask conditioning can stabilize long-horizon behavior.

IV. Results

In this section, we present experimental results evaluating the proposed framework in the RoboTwin simulation environment, which is designed for long-horizon robot manipulation tasks with diverse initial conditions.

We consider two representative manipulation tasks:

- **Task 1**: Moving objects between multiple target locations (*blocks ranking rgb*)
- **Task 2**: Grasping and placing objects into a cabinet (*put object cabinet*)



(a) Task 1: Before (b) Task 1: After

(c) Task 2: Before (d) Task 2: After

Fig. 2: Illustration of task execution processes. (a) and (b) show the initial and final states of Task 1, while (c) and (d) correspond to Task 2.

For each task, we evaluate performance over 500 independent test episodes with randomized initial states. The primary evaluation metric is task success rate, defined as the percentage of episodes in which the task is completed within the time limit.

A. Main Results

We compare the proposed split-rubric framework with a standard end-to-end baseline across both tasks. Specifically, we evaluate the $\pi_0 - base$ model trained with the original RoboTwin prompt, and its counterpart $\pi_0 - split$, which uses oracle subtask-level prompts. We further report results for the stronger $\pi_{0.5} - base$ model when available.

Tables I and II summarize the success rates under different rubric settings.

TABLE I: Success rates under different rubrics on *blocks ranking rgb*.

Model	Rubric	
	Baseline	Split
π_0-base	45.4%	47.6%
π_0-split	35.6%	53.8%
$\pi_{0.5}$-base	68.3%	74.6%

On Task 1, the split rubric leads to a noticeable improvement for the π_0-split model, increasing the success rate from 45.4% to 53.8%. In contrast, the π_0-base model exhibits only marginal changes. This result suggests that explicit task

TABLE II: Success rates under different rubrics on *put object cabinet*.

Model	Rubric	
	Baseline	Split
π_0-base	31.2%	31.0%
π_0-split	28.0%	27.8%

decomposition can be beneficial for long-horizon manipulation tasks that involve multiple loosely coupled subtasks.

One possible explanation is that conditioning the policy on subtask-level prompts encourages the learning of more localized action patterns, reducing reliance on a single fixed execution trajectory. As a result, the policy may exhibit greater tolerance to deviations from the expected sequence of actions. In comparison, the end-to-end baseline primarily relies on trajectory-level imitation, which can be brittle when execution errors accumulate.

On Task 2, however, the split rubric does not yield improvements and results in a slight degradation. This outcome indicates that task decomposition alone may be insufficient for tasks that require precise coordination and limited recovery opportunities. A more detailed investigation into data coverage and recovery mechanisms is left for future work.

B. Prompt Sensitivity Analysis

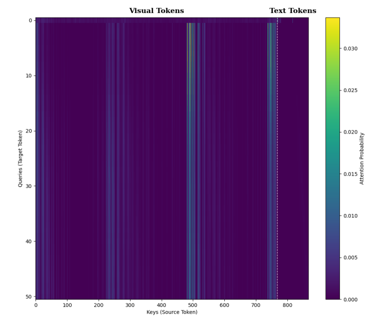
To further analyze the differing behaviors of π_0 and $\pi_{0.5}$, we visualize attention distributions from the final transformer layer, as shown in Fig. 3.

In π_0, attention is predominantly concentrated on visual tokens, with relatively weak attention to textual inputs across both averaged and individual heads. This pattern suggests that action generation in π_0 is largely driven by visual observations, with limited sensitivity to changes in language prompts. This observation is consistent with the modest performance differences observed under different prompting strategies.

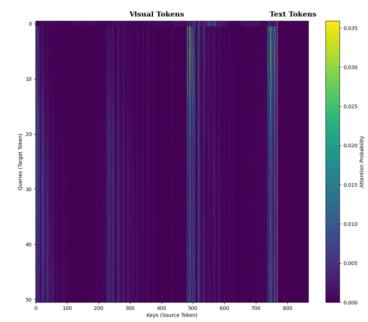
In contrast, $\pi_{0.5}$ allocates a larger portion of attention to textual tokens, indicating a stronger dependence on language conditioning. This difference aligns with the improved performance of $\pi_{0.5}$ under the split rubric, where subtask-level prompts provide more structured guidance. While attention visualizations are descriptive and do not establish causality, the observed patterns are consistent with the relative prompt sensitivity of the two models.

C. Case Studies

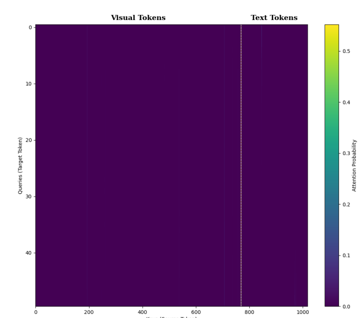
a) Subtask-level adaptation in Task 1: In Task 1, the split model occasionally exhibits adaptive behaviors when partial failures occur. For example, if an early object placement is unsuccessful, the policy may proceed with subsequent subtasks and later return to reattempt the failed one. Such behavior is rarely observed in the end-to-end baseline and suggests reduced dependence on a fixed global trajectory.



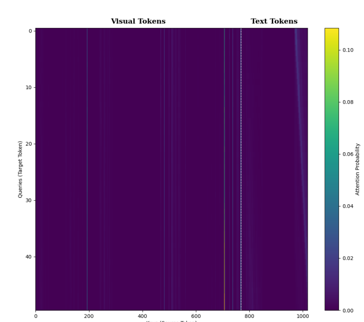
(a) π_0: all-head average



(b) π_0: head 6



(c) $\pi_{0.5}$: all-head average



(d) $\pi_{0.5}$: head 6

Fig. 3: Visualization of attention distributions in the final transformer layer. (a) Average attention over all heads in the final layer of the π_0 model. (b) Attention map of Head 6 in the final layer of the π_0 model. (c) Average attention over all heads in the final layer of the $\pi_{0.5}$ model. (d) Attention map of Head 6 in the final layer of the $\pi_{0.5}$ model.

b) Hand-role consistency in Task 2: In Task 2, the split model shows more consistent hand selection during grasping. Compared to the baseline, which occasionally selects an inappropriate hand, the split model exhibits fewer such errors. Although this improvement does not translate into higher overall success rates, it indicates that subtask-aware prompting can still enhance fine-grained action correctness.

Overall, these observations suggest that explicit task decomposition can improve execution robustness in certain long-horizon settings by reducing reliance on trajectory-level imitation, even without explicit recovery mechanisms.

V. Discussion

This section discusses the empirical findings and their implications for hierarchical VLA formulations.

A. Impact of Task Decomposition

The results indicate that explicit task decomposition can improve execution reliability for certain long-horizon manipulation tasks. Rather than serving as a universally applicable solution, task decomposition appears most effective when the task admits a clear temporal structure and loosely coupled subtasks.

Such decomposition is particularly relevant for tasks with clear stage boundaries, where errors in early stages do not necessarily preclude successful completion of later stages. By contrast, for tasks with tightly coupled action sequences, the benefits of decomposition are less evident, as observed in Task 2. These observations indicate that task decomposition is not universally beneficial, but its effectiveness depends on the structure and temporal abstraction level of the task.

B. Effect of Subtask-Specific Finetuning

Subtask-specific finetuning further refines the benefits of task decomposition by aligning the policy with the distribution of subtask-level prompts and observations. In our experiments, finetuning π_0 with subtask-specific prompts improves performance under the split rubric, while introducing a mismatch under the baseline rubric. This behavior highlights that the observed gains primarily arise from structured prompting and localized supervision, rather than from generic improvements in model capacity.

The results suggest that subtask-specific finetuning encourages the model to learn more reusable and atomic behaviors, which may contribute to increased execution stability. However, the magnitude of improvement is constrained by the model's intrinsic sensitivity to language inputs, as indicated by the attention analysis. Models with stronger prompt conditioning, such as $\pi_{0.5}$, appear better positioned to benefit from hierarchical prompting strategies.

C. Comparison with End-to-End Baselines

Compared to the end-to-end π_0 baseline, the proposed framework achieves higher success rates on tasks where hierarchical structure is well aligned with the task dynamics. The baseline model, trained via trajectory-level imitation, is more sensitive to deviations from the training distribution and often fails to recover from intermediate errors. In contrast, the hierarchical approach reduces reliance on a fixed global trajectory by conditioning action generation on the current subtask.

Nevertheless, the results also indicate that hierarchical prompting alone does not guarantee improved performance. In Task 2, where fine-grained coordination between successive actions is required, task decomposition yields limited benefits. This comparison underscores that hierarchical structure should be applied selectively, based on the characteristics of the target task.

D. Limitations

Several limitations of the current study should be noted. First, task decomposition relies on oracle-provided subtasks, which limits the autonomy of the framework. While this setting allows for controlled analysis of hierarchical prompting, it does not address how subtasks should be generated in practice. Second, the evaluation is conducted on a limited set of tasks in simulation, which constrains the scope of the conclusions. Finally, the framework does not incorporate explicit mechanisms for error detection or recovery, which may be necessary to fully exploit the potential of hierarchical execution in more complex scenarios.

VI. Conclusion

This work presents an empirical analysis of hierarchical task decomposition and subtask-specific finetuning in visual-language-action models for long-horizon robot manipulation. By adopting an oracle-based hierarchical prompting formulation, we analyze how explicit task structure influences execution robustness under a fixed policy architecture.

The results suggest that hierarchical conditioning can reduce reliance on trajectory-level imitation and improve execution consistency for tasks with meaningful subtask abstraction. Attention-based analysis further indicates that the effectiveness of such prompting strategies depends on a model's sensitivity to language inputs.

Future work may explore automatic subtask generation and integrate explicit recovery mechanisms to further improve robustness in more complex and realistic environments.

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